

Written Homework 2

Problem Solving, Math 1000 – Summer 2016

Due Friday, June 11th

Problem 1 Write each of the following percentages as a decimal:

(a) 5.49%

(e) 0.03%

(b) $\frac{4}{5}\%$

(f) 3.25%

(c) $\frac{1}{10}\%$

(d) 0.64%

(g) 0.006%

Problem 2 A 1800 piece puzzle is missing 16% of its pieces. How many pieces are left?

Problem 3 Leo paid \$15,300 in tuition last year. If his tuition is \$18,254 this semester, by what percent did his tuition increase?

Problem 4 Home values in a town have declined 17% per year for each of the past four years. What was the total percentage decrease in home values during the four-year period?

Problem 5 An electronics store purchases a certain video game from the manufacturer for \$18 per unit. The store sells the game at a mark-up of 120% over the purchase price. You found a coupon, offered in the local newspaper, which will reduce the price by 20%. How much money would the store profit if you purchased the video game using the coupon? Give your answer as a decimal rounded to the nearest hundredth.

Problem 6 When you were born your grandmother purchased a low-risk government savings bond for \$1250. The bond pays 5.25% annual simple interest. Determine the future value of the bond after seven years. Give your answer as a decimal rounded to the nearest hundredth.

Problem 7 The local Rent-to-Own Store offers you the following plan to purchase a washing machine. The washing machine retails for \$850, but you will have three years to pay it off using a simple interest loan. If you pay back \$1400, then what was the APR on this loan? Give your answer as a percentage rounded to the nearest tenth of a percent.

Problem 8 You purchase a new car after graduating from college. Each year the assessed value of the car decreases by 10%. After five years, what is the total percentage decrease on the value of the car? Give your answer as a percentage rounded to the nearest tenth of a percent.